Experiment 5: Transistor Small-Signal Amplifiers

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EEE3307-C0013: Electronics I

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5.A Objective

The goal of this experiment is to study small-signal transistor amplifiers built around the 2N2222 BJT. A common-emitter (CE) amplifier with and without an emitter bypass capacitor and a common-collector (emitter follower) amplifier are designed, simulated, and built. For each circuit the Q-point, maximum unclipped output, small-signal voltage gains $A_{\text{vi}} = \frac{v_o}{v_i}$ and $A_{\text{vs}} = \frac{v_o}{v_s}$, and the input and output resistances are calculated, then compared against simulation and laboratory measurements.

5.B Pre-Lab

Given values: $V_{\text{CC}} = 12V$, $R_C = 6.2k\Omega$, $R_E = 1.8k\Omega$, $R_L = 2.2k\Omega$, $R_S = 1.8k\Omega$, $C_E = 100\mu F$, $C_B = C_L = 10\mu F$. For the 2N2222 the lab manual gives $\beta \approx 150$ and $V_A \approx 100V$. Throughout, $V_{\text{BE}} = 0.7V$, $V_T = kT/q \approx 26mV$ and $V_{\text{CE(sat)}} \approx 0.2V$ are assumed.

5.B.1 Common-Emitter Amplifier

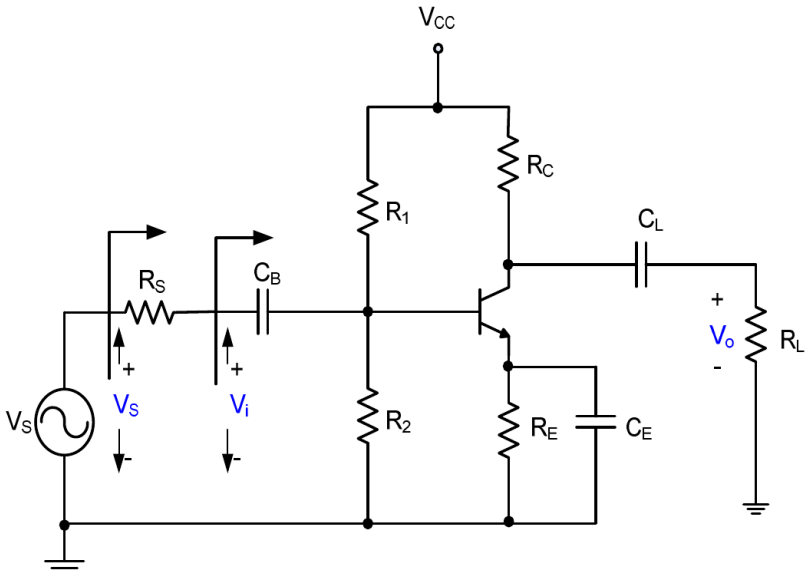


Figure 1: Common-emitter amplifier

5.B.1.a Calculate R_1 and R_2

Targeting $I_{\text{CQ}} \approx 1mA$

$$R_{\text{TH}} = .1 \times (1+\beta) \times R_E \\ = .1 \times 151 \times 1.8k\Omega \approx 27.2k\Omega \quad \text{Eq 1}$$

$$I_B = \frac{1mA}{150} = 6.67\mu A \\ I_E = I_B \times (1+\beta) = 1.007mA \quad \text{Eq 2}$$

$$V_{\text{TH}} = V_{\text{BE}} + I_E \times R_E + I_B \times R_{\text{TH}} \\ = .7 + 1.81 + .18 = 2.69V \quad \text{Eq 3}$$

$$V_{\text{TH}} = V_{\text{CC}} \times \frac{R_2}{R_1 + R_2} \\ R_1 = 121.249k\Omega \approx 120k\Omega \\ R_2 = 35.033k\Omega \approx 36k\Omega \quad \text{Eq 4}$$

5.B.1.b Compute Q-point and max unclipped output voltage

$$R_{\text{TH}} = R_1 \parallel R_2 = 120k\Omega \parallel 36k\Omega = 27.69k\Omega$$
$$V_{\text{TH}} = 12 \times \frac{36k}{120k + 36k} = 2.769V$$
Eq 5

$$I_{\text{BQ}} = \frac{V_{\text{TH}} - V_{\text{BE}}}{R_{\text{TH}} + (1+\beta)R_E} = \frac{2.769 - .7}{27.69k + 151 \times 1.8k} = 6.91\mu A$$
$$I_{\text{CQ}} = \beta I_{\text{BQ}} = 1.036mA, \quad I_{\text{EQ}} = (1+\beta)I_{\text{BQ}} = 1.043mA$$
Eq 6

$$V_{\text{CEQ}} = V_{\text{CC}} - I_{\text{CQ}}R_C - I_{\text{EQ}}R_E$$
$$= 12 - 1.036m \times 6.2k - 1.043m \times 1.8k = 3.70V$$
Eq 7

The Q-point is therefore $\left(I_{\text{CQ}}, V_{\text{CEQ}}\right) = (1.04mA, 3.70V)$.

$$R_{\text{ac}} = R_C \parallel R_L = 6.2k \parallel 2.2k = 1.624k\Omega$$
Eq 8

to find the maximum unclipped output

$$\text{Saturation limit:} \quad V_{\text{CEQ}} - V_{\text{CE(sat)}} = 3.70 - 0.2 = 3.50V$$
$$\text{Cutoff limit:} \quad I_{\text{CQ}}R_{\text{ac}} = 1.036m \times 1.624k = 1.68V$$
Eq 9

The cutoff limit is smaller, so the maximum unclipped output is $\approx 3.37V_{\text{pp}}$.

5.B.1.c Compute the small-signal voltage gain $A_{\text{vi}} = \frac{v_o}{v_i}$ and $A_{\text{vs}} = \frac{v_o}{v_s}$

$$g_m = \frac{I_{\text{CQ}}}{V_T} = \frac{1.036m}{26m} = 39.86mA/V$$
$$r_\pi = \frac{\beta}{g_m} = \frac{150}{39.86m} = 3.763k\Omega$$
$$r_o = \frac{V_A}{I_{\text{CQ}}} = \frac{100}{1.036m} = 96.5k\Omega$$
Eq 10

$$A_{\text{vi}} = -g_m(r_o \parallel R_C \parallel R_L) = -39.86m \times 1.597k = -63.7$$
$$A_{\text{vs}} = A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = -63.7 \times \frac{3.313k}{3.313k + 1.8k} = -41.2$$
Eq 11

5.B.1.d Compute the input and output resistances

$$R_i = R_1 \parallel R_2 \parallel r_\pi = 27.69k \parallel 3.763k = 3.31k\Omega$$
$$R_o = R_C \parallel r_o = 6.2k \parallel 96.5k = 5.83k\Omega$$
Eq 12

5.B.2 Common-Emitter Amplifier without C_E

5.B.2.a Compute Q-point and max unclipped output voltage

C_E is an open circuit at DC, so removing it does not change the bias: the Q-point is still $(I_{\text{CQ}}, V_{\text{CEQ}}) = (1.04mA, 3.70V)$.

$$R_{\text{ac}} = R_C \parallel R_L + R_E = 1.624k + 1.8k = 3.424k\Omega \quad \text{Eq 13}$$

to find max swing

$$\begin{aligned} \text{Saturation:} \quad & i_{\text{c,max}} = \frac{V_{\text{CEQ}} - V_{\text{CE(sat)}}}{R_{\text{ac}}} = \frac{3.70 - 0.2}{3.424k} = 1.021mA \rightarrow v_o = 1.021m \times 1.624k = 1.66V \\ \text{Cutoff:} \quad & i_{\text{c,max}} = I_{\text{CQ}} = 1.036mA \rightarrow v_o = 1.036m \times 1.624k = 1.68V \end{aligned} \quad \text{Eq 14}$$

Saturation is reached first, so the maximum unclipped output is $\approx 3.32V_{\text{pp}}$.

5.B.2.b Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

The resistance looking into the base now includes the reflected emitter resistor:

$$R_{\text{ib}} = r_\pi + (1+\beta)R_E = 3.763k + 151 \times 1.8k = 275.6k\Omega \quad \text{Eq 15}$$

$$\begin{aligned} A_{\text{vi}} &= \frac{-\beta(R_C \parallel R_L)}{r_\pi + (1+\beta)R_E} = \frac{-150 \times 1.624k}{275.6k} = -0.884 \\ A_{\text{vs}} &= A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = -0.884 \times \frac{25.16k}{25.16k + 1.8k} = -0.825 \end{aligned} \quad \text{Eq 16}$$

5.B.2.c Compute the input and output resistances

$$\begin{aligned} R_i &= R_1 \parallel R_2 \parallel [r_\pi + (1+\beta)R_E] = 27.69k \parallel 275.6k = 25.2k\Omega \\ R_o &\approx R_C = 6.2k\Omega \end{aligned} \quad \text{Eq 17}$$

5.B.3 Common-Collector (Emitter Follower) Amplifier

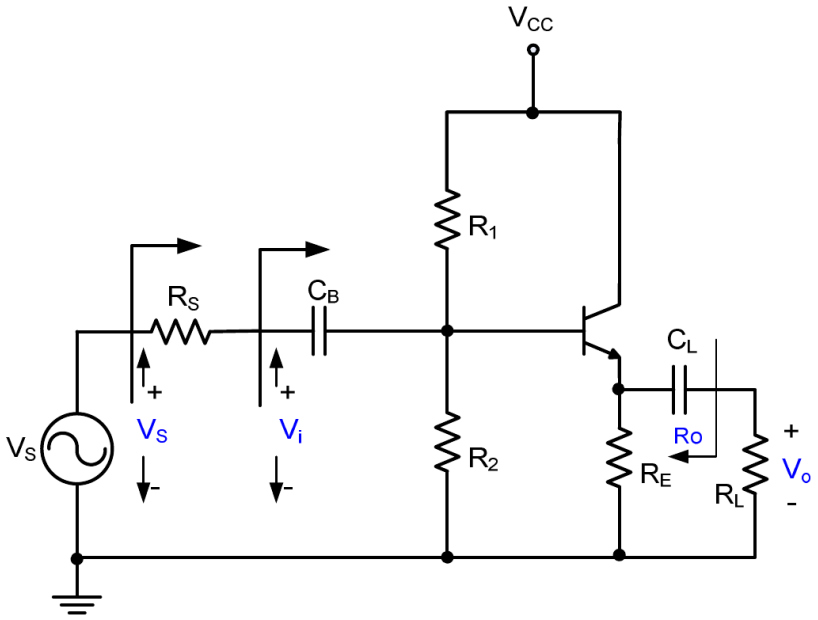


Figure 2: Emitter follower

5.B.3.a Calculate R_1 and R_2

$$R_{\text{TH}} = .1 \times (1+\beta) \times R_E = .1 \times 151 \times 1.8k\Omega \approx 27.2k\Omega$$
$$V_{\text{TH}} = V_{\text{BE}} + I_E \times R_E + I_B \times R_{\text{TH}} = .7 + 1.81 + .18 = 2.69V \quad \text{Eq 18}$$

$$R_1 = R_{\text{TH}} \times \frac{V_{\text{CC}}}{V_{\text{TH}}} = 121.2k\Omega \approx 120k\Omega$$
$$R_2 = R_{\text{TH}} \times \frac{V_{\text{CC}}}{V_{\text{CC}} - V_{\text{TH}}} = 35.0k\Omega \approx 36k\Omega \quad \text{Eq 19}$$

The same $R_1 = 120k\Omega$ and $R_2 = 36k\Omega$ were therefore used for both circuits.

5.B.3.b Compute Q-point and max unclipped output voltage

$$I_{\text{BQ}} = 6.91\mu A, \quad I_{\text{CQ}} = 1.036mA, \quad I_{\text{EQ}} = 1.043mA$$
$$V_E = I_{\text{EQ}} R_E = 1.043m \times 1.8k = 1.88V$$
$$V_{\text{CEQ}} = V_{\text{CC}} - I_{\text{EQ}} R_E = 12 - 1.88 = 10.12V \quad \text{Eq 20}$$

The Q-point is $\left(I_{\text{CQ}}, V_{\text{CEQ}}\right) = (1.04mA, 10.1V)$.

$$\text{Saturation limit:} \quad V_{\text{CEQ}} - V_{\text{CE(sat)}} = 10.12 - 0.2 = 9.92V$$
$$\text{Cutoff limit:} \quad I_{\text{EQ}} R_{\text{ac}} = 1.043m \times 990 = 1.03V \quad \text{Eq 21}$$

The maximum unclipped output is $\approx 2.07V_{\text{pp}}$, limited by cutoff.

5.B.3.c Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

The effective emitter load is $R'_E = R_E \parallel R_L \parallel r_o = 1.8k \parallel 2.2k \parallel 96.5k = 980\Omega$.

$$A_{\text{vi}} = \frac{(1+\beta)R'_E}{r_\pi + (1+\beta)R'_E} = \frac{151 \times 980}{3.763k + 151 \times 980} = \frac{147.97k}{151.73k} = 0.975$$
$$A_{\text{vs}} = A_{\text{vi}} \times \frac{R_i}{R_i + R_S} = 0.975 \times \frac{23.42k}{23.42k + 1.8k} = 0.906$$

Eq 22

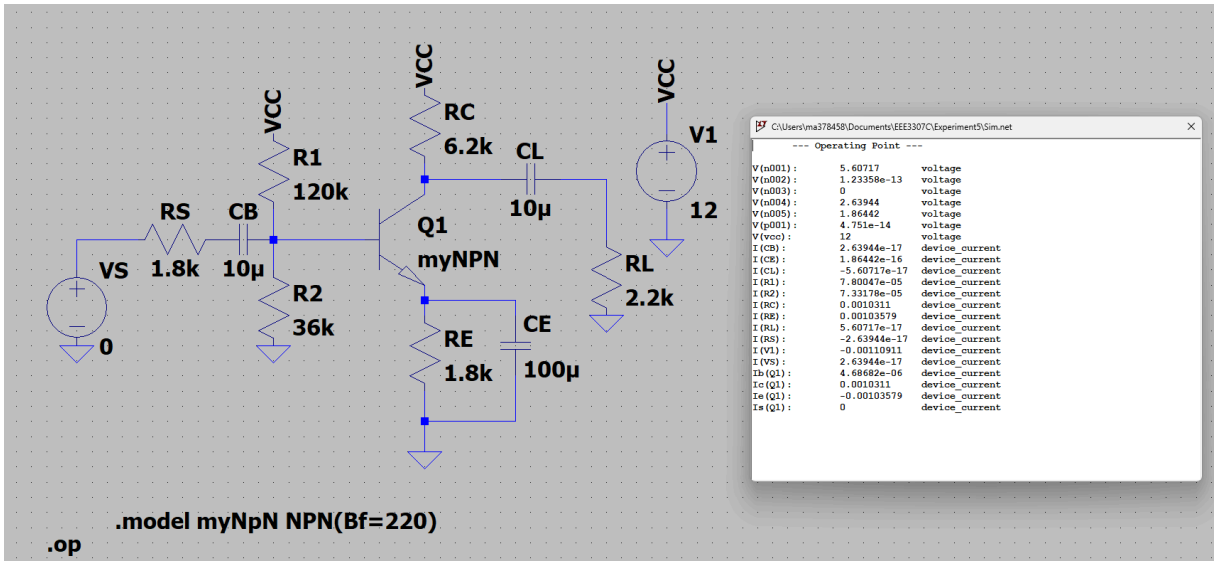
5.B.3.d Compute the input and output resistances

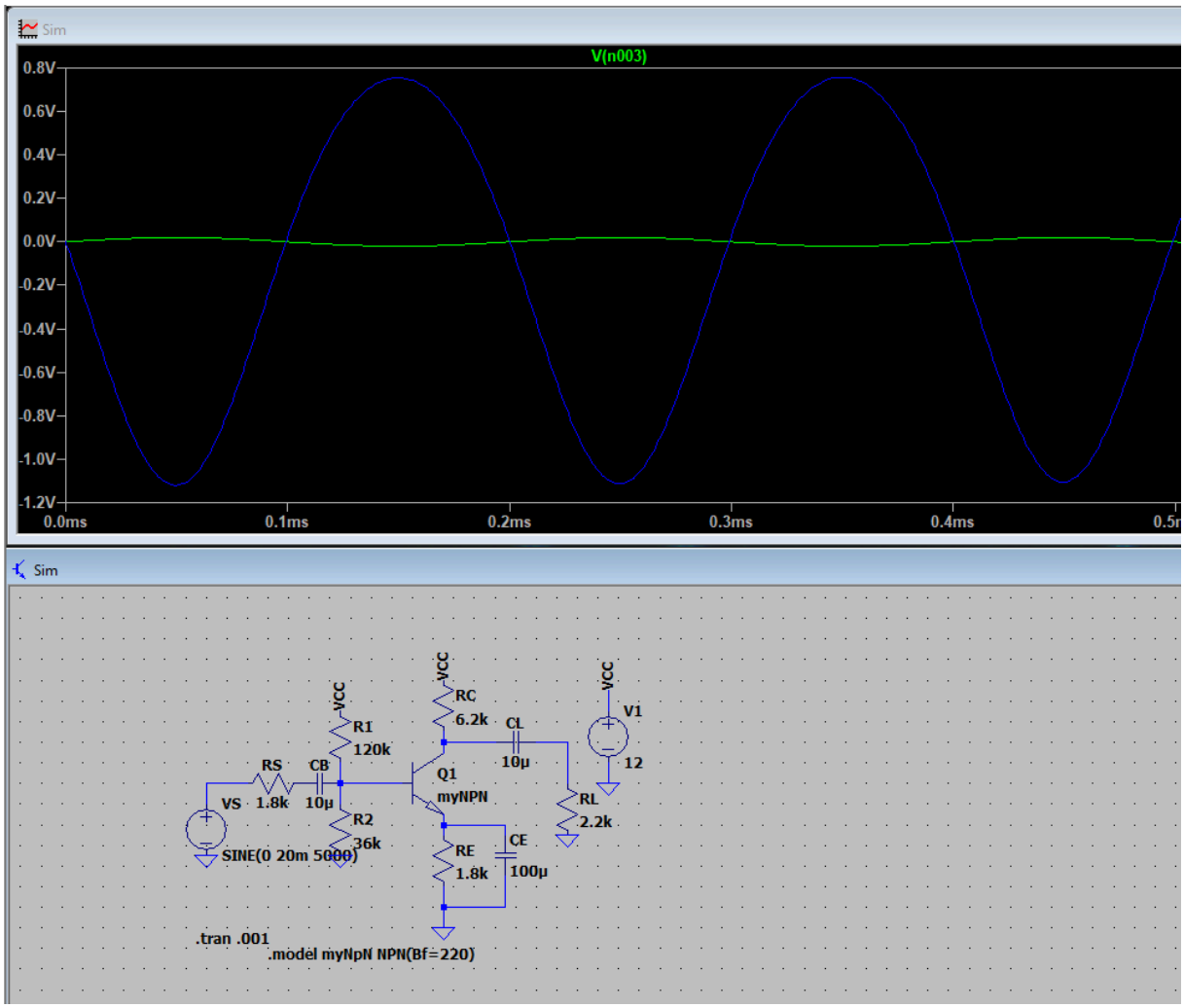
$$R_i = R_1 \parallel R_2 \parallel [r_\pi + (1+\beta)R'_E] = 27.69k \parallel 151.7k = 23.4k\Omega$$
$$R_o = \frac{r_\pi + R_1 \parallel R_2 \parallel R_S}{1+\beta} \parallel R_E \parallel r_o$$
$$= \frac{3.763k + 1.690k}{151} \parallel 1.8k \parallel 96.5k = 36.1 \parallel 1.8k \parallel 96.5k = 35.4\Omega$$

Eq 23

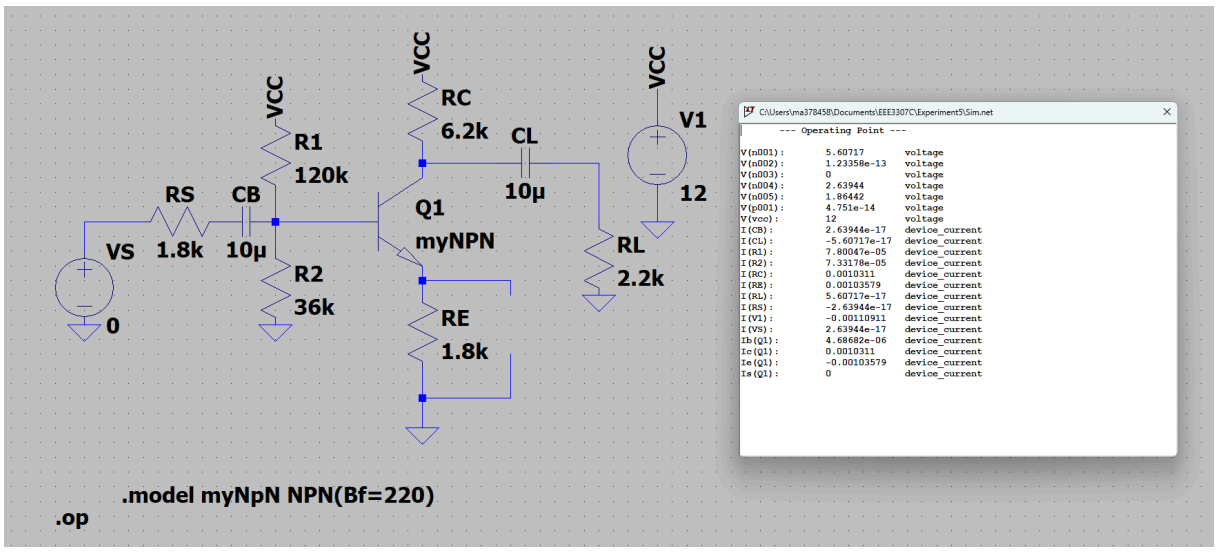
5.C Simulation

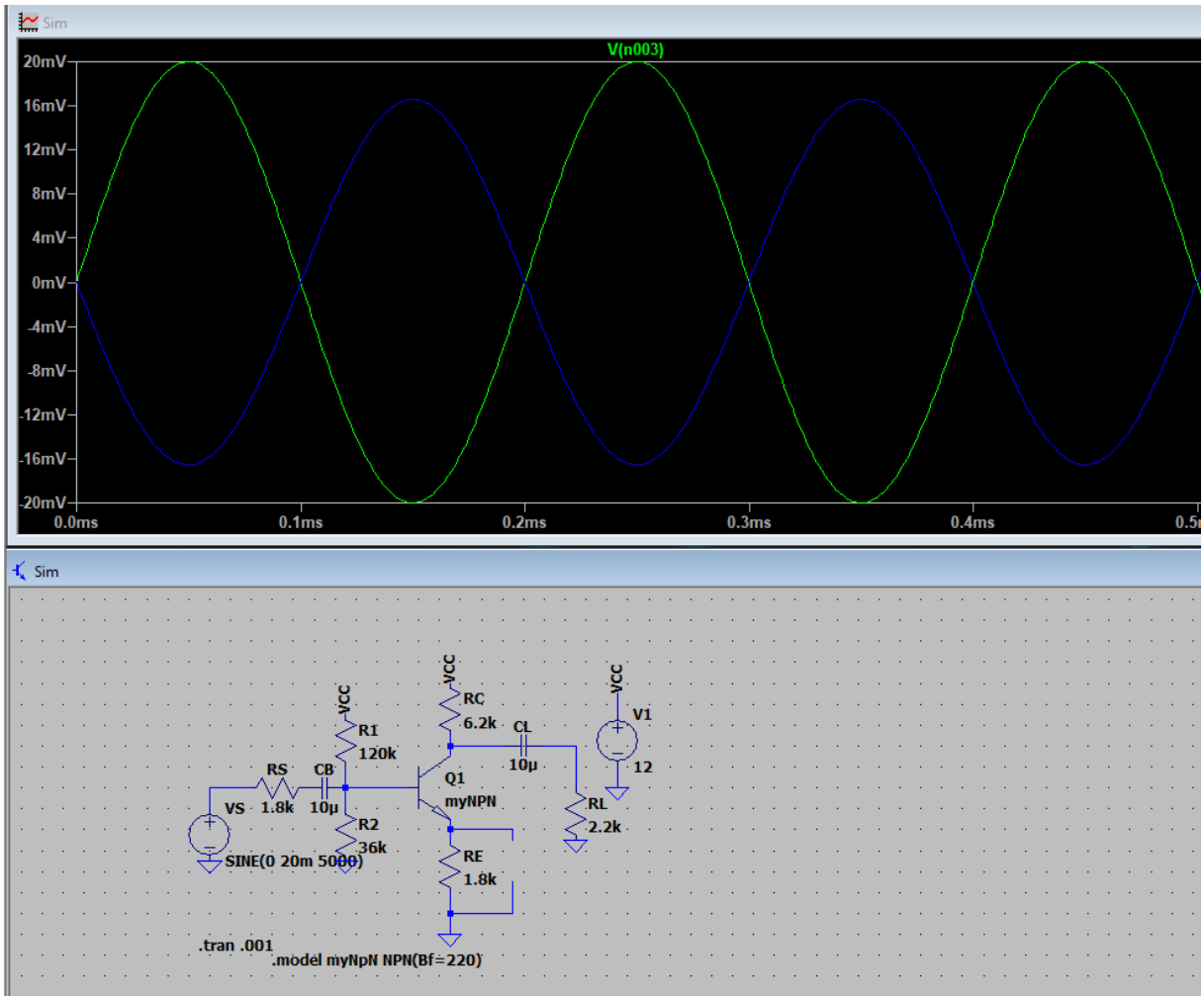
With CE



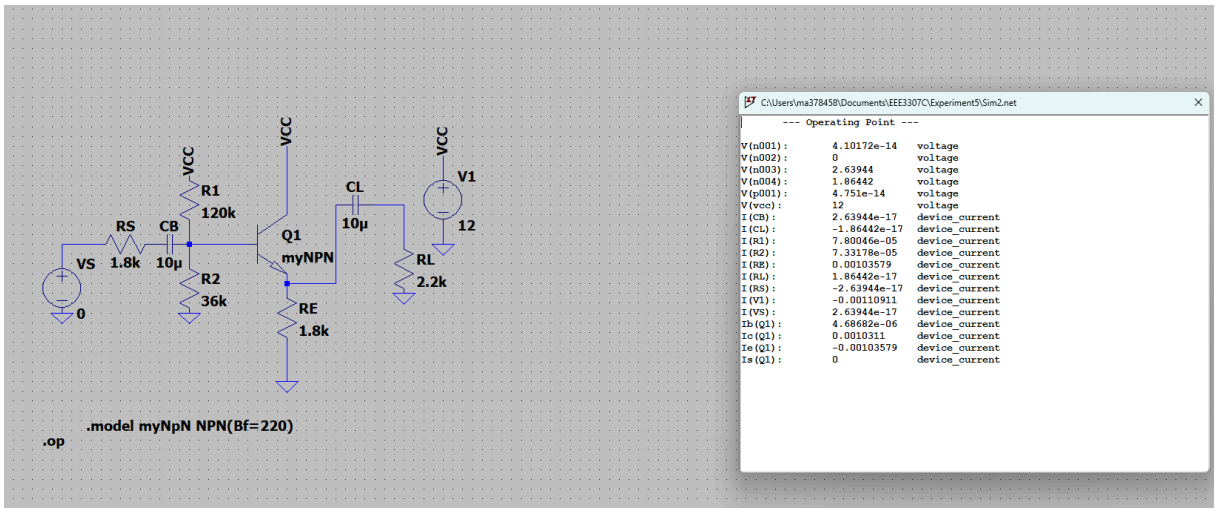

Sim
V(n003)
0.8V
0.6V
0.4V
0.2V
0.0V
-0.2V
-0.4V
-0.6V
-0.8V
-1.0V
-1.2V
0.0ms
0.1ms
0.2ms
0.3ms
0.4ms
Sim
VCC
R1
120k
RC
6.2k
CL
10µ
V1
12
RS
CB
Q1
myNPN
VS
1.8k
10µ
R2
36k
RL
2.2k
RE
1.8k
CE
100µ
SINE(0 20m 5000)
.tran .001
.model myNpN NPN(Bf=220)

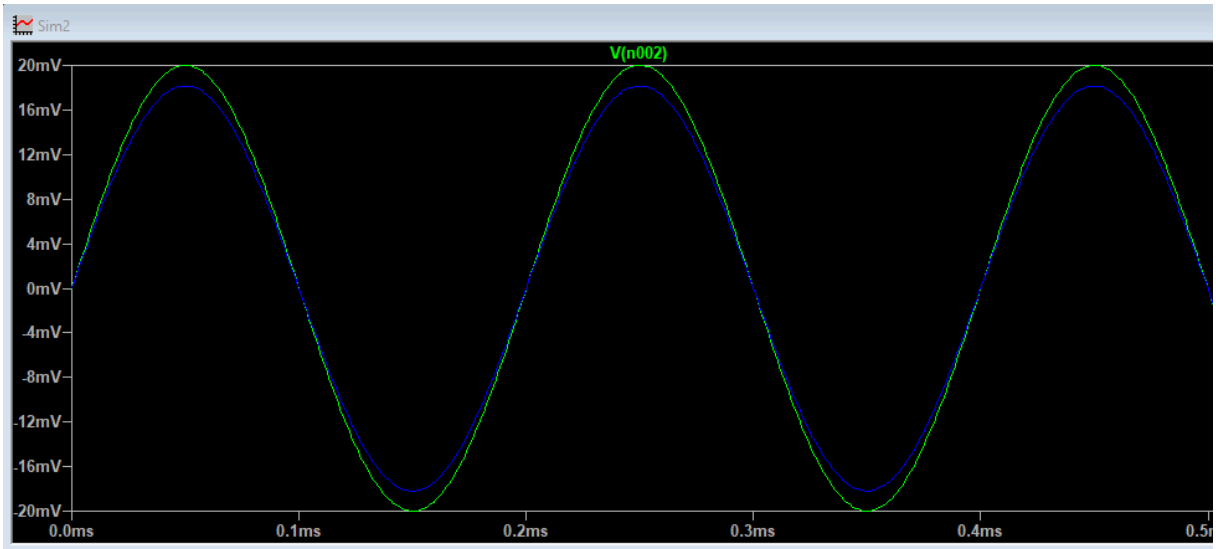
Without CE

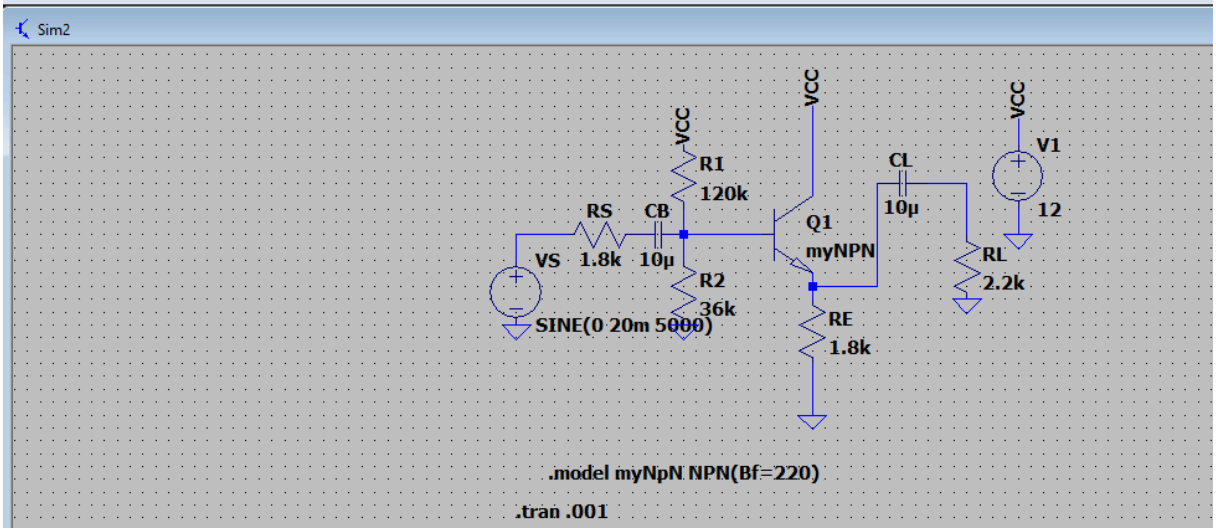

VCC
R1
120k
RC
6.2k
CL
10µ
VCC
V1
12
RS
CB
Q1
myNPN
VS
1.8k
10µ
R2
36k
RL
2.2k
RE
1.8k
0
.model myNpN NPN(Bf=220)
.op
C:\Users\ma378458\Documents\EEE3307C\Experiment5\Sim.net
--- Operating Point ---
V(n001): 5.60717 voltage
V(n002): 1.23358e-13 voltage
V(n003): 0 voltage
V(n004): 2.63944 voltage
V(n005): 1.86442 voltage
V(p001): 4.751e-14 voltage
V(vcc): 12 voltage
I(CB): 2.63944e-17 device_current
I(CL): -5.60717e-17 device_current
I(R1): 7.80047e-05 device_current
I(R2): 7.33178e-05 device_current
I(RC): 0.0010311 device_current
I(RE): 0.00103579 device_current
I(RL): 5.60717e-17 device_current
I(RS): -2.63944e-17 device_current
I(V1): -0.00110911 device_current
I(VS): 2.63944e-17 device_current
Ib(Q1): 4.68682e-06 device_current
Ic(Q1): 0.0010311 device_current
Ie(Q1): -0.00103579 device_current
Is(Q1): 0 device_current



Emitter follower







5.D Experimental Procedure

The circuits of Figure 1 and Figure 2 were built on a breadboard with $R_1 = 120k\Omega$, $R_2 = 36k\Omega$, $C_B = C_L = 10\mu F$ and $C_E = 100\mu F$, powered from a 12 V supply.

5.D.1 Common-Emitter Amplifier

Q-point: $I_{CQ} = 1.11mA$, $V_{CEQ} = 2.37V$. **Gains:** $v_s = 7.0mV_{rms}$, $v_i = 5.0mV_{rms}$, $v_o = 308mV_{rms}$

$$A_{vs} = -\frac{308m}{7.0m} = -44.0, \quad A_{vi} = -\frac{308m}{5.0m} = -61.6 \qquad \text{Eq 24}$$

The negative sign comes from the oscilloscope captures (Figure 3, Figure 4), where the output is 180° out of phase with the input.

Input resistance: $V_{R_S} = 2.3mV_{rms}$

$$i_s = \frac{2.3mV}{1.8k\Omega} = 1.28\mu A, \quad R_i = \frac{v_i}{i_s} = \frac{5.0mV}{1.28\mu A} = 3.91k\Omega \qquad \text{Eq 25}$$

Output resistance: $v_L = 308mV_{rms}$ (with R_L), $v_o = 960mV_{rms}$ (without R_L)

$$R_o = R_L \times \frac{v_o - v_L}{v_L} = 2.2k \times \frac{960 - 308}{308} = 4.66k\Omega \qquad \text{Eq 26}$$

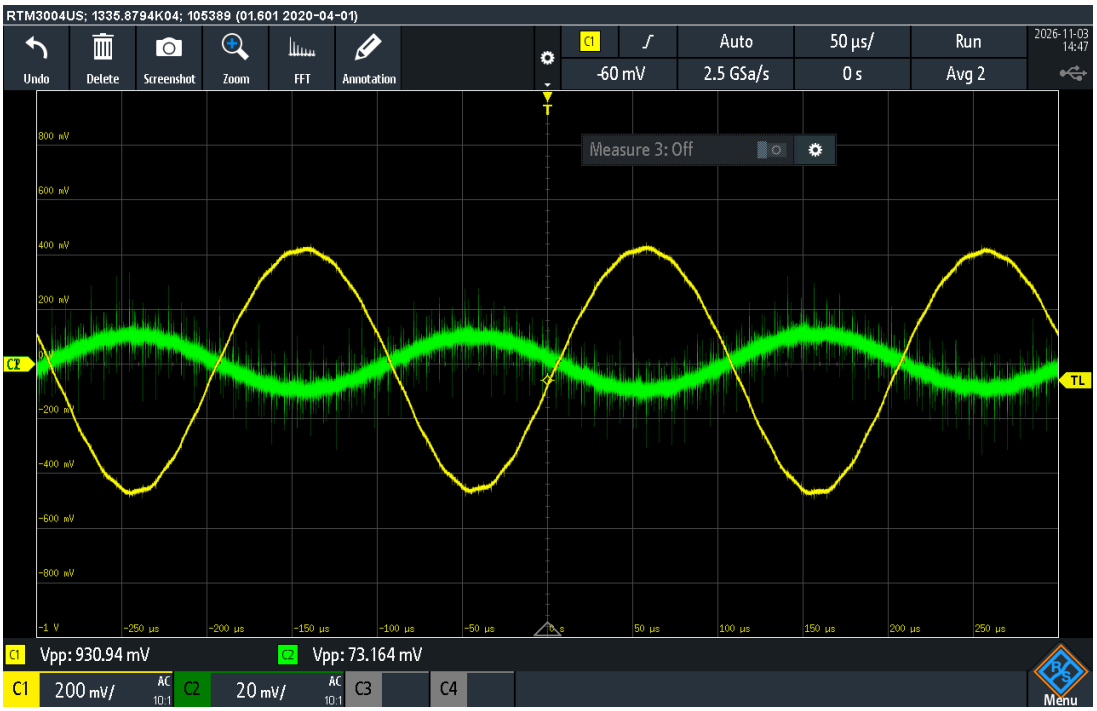


Figure 3: CE amplifier, v_s (CH2, green, 20 mV/div) and v_o (CH1, yellow, 200 mV/div) at 5 kHz

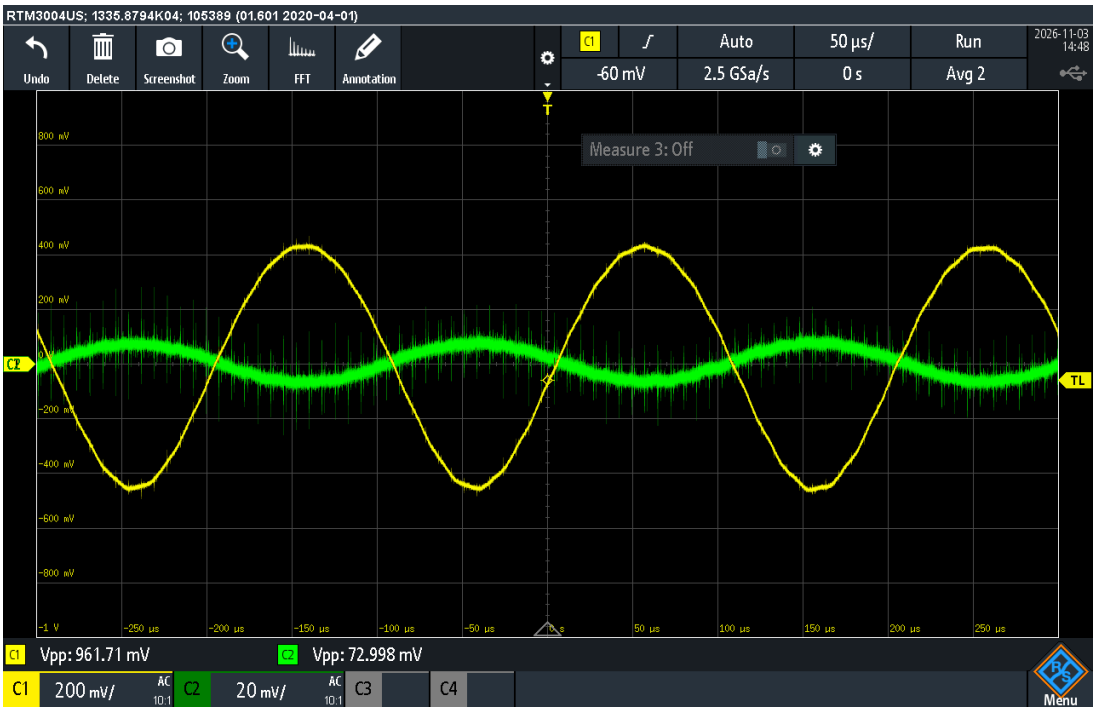


Figure 4: CE amplifier, v_i (CH2, green, 20 mV/div) and v_o (CH1, yellow, 200 mV/div) at 5 kHz

5.D.2 Common-Collector (Emitter Follower) Amplifier

Gains: $v_s = 7.0mV_{\text{rms}}$, $v_i = 6.6mV_{\text{rms}}$, $v_o = 6.4mV_{\text{rms}}$

$$A_{\text{vs}} = \frac{6.4m}{7.0m} = 0.914, \quad A_{\text{vi}} = \frac{6.4m}{6.6m} = 0.970 \qquad \text{Eq 27}$$

Input resistance: $V_{R_S} = 1.3mV_{\text{rms}}$

$$i_s = \frac{1.3mV}{1.8k\Omega} = 0.722\mu A, \quad R_i = \frac{v_i}{i_s} = \frac{6.6mV}{0.722\mu A} = 9.14k\Omega \qquad \text{Eq 28}$$

Output resistance: $v_L = 6.4mV_{\text{rms}}$ (with R_L), $v_o = 6.5mV_{\text{rms}}$ (without R_L)

$$R_o = R_L \times \frac{v_o - v_L}{v_L} = 2.2k \times \frac{6.5 - 6.4}{6.4} = 34.4\Omega \qquad \text{Eq 29}$$